Build a cable. Make a connection.

I can explain a small network, put eight wires in order, and use a test to improve my work.

7 tables | 2 class visits | Same task. Shared tools.

What is a network?

A **network** connects devices so they can share data. The **internet** connects many networks. A small network can work without the internet.

Devices (Computer + printer) ↔ **Switch** (Connects local devices) ↔ **Router** (Connects networks) ↔ **Internet** (Many networks)

Try a paper network

Draw two devices and a switch. Trace a message from one device to the other. Add a router and an outside network. Trace a message that leaves your network.

Talk: Does every message need the internet?

What we will make

A short **Ethernet cable** with an RJ45 plug on each end. Inside are four twisted pairs: eight wires in all.

Our cable stays on the table. We test it with a separate battery-powered cable tester.

Everyone has a turn

1 · Builder
Arrange and insert wires.

2 · Checker
Read the color order aloud.

3 · Tester
Read both sets of lights.

4 · Recorder
Draw the network and record results.

Work with up to four people. Move jobs one place after each end or test. In a smaller group, share jobs. Everyone arranges wires, reads a check and explains a result. Ask for help with grip or close-up work.

Our tool rule

Wait for the demonstration and approval before crimping. Keep fingers clear of blades and the crimp opening. Use only the loose practice cable. Never plug it into a wall jack, school device, switch, router or PoE equipment.

Same order. Both ends.

Use **T568B at both ends**. A plug goes on only once. Check before you crimp.

Plug tip points away from you ↑



Look down at the gold contacts. The clip is underneath. Cable comes toward you. Read left to right, 1 to 8. Do not flip the plug when comparing it with this picture.

1. **Prepare.** Use the teacher-prepared loose cable. Slide a boot onto the cable before adding a plug, with its wide opening toward the end. The teacher shows how to remove only the outer jacket. Do not strip the eight little wires.
2. **Sort.** Keep each pair twisted as close to the jacket as you can. Open only enough to line up the wires. Arrange all eight in the numbered order above. A partner checks.
3. **Insert.** With approved pass-through plugs, guide one wire into each channel until all eight tips come through the front. Push the jacket inside under the strain-relief tab. Stop if a wire will not fit.
4. **Stop and show.** Check colors, all eight channels and jacket position. Get the teacher's approval. Crimp once with the matching pass-through tool. Its cutter must trim the tips flush; no copper should stick out. Slide the boot forward.
5. **Swap jobs.** Make the second end in the same T568B order. Keep your first end for comparison. Show both ends before the second crimp.
6. **Test.** Click one end into the tester and the other into its remote. Switch to the demonstrated slow test. Match the main and remote lights: **1→1, 2→2, 3→3, 4→4, 5→5, 6→6, 7→7, 8→8.** Watch two full runs. Switch it off when done.

Finish check

- □ Both ends match T568B.
- □ Eight positions light in matching order.
- □ Jackets are held; tips are flush.
- □ I can explain what I checked.

If it does not pass

Record what happened. Check the colors and the plug first. Ask before cutting off a bad end. Use a **new plug** for the repair, then test again. A useful failed test is learning.

A basic tester checks connections. It does not prove internet access, speed or every wiring fault. The teacher also inspects the pairs and both plugs. On our unshielded cable, a ground or shield light may stay off.

Seven tables. One shared challenge.

Two 35-minute visits per class. Repeat each visit across the five-day rotation before the same class returns. Keep the current annual playbook position unchanged.

At every table

- One compatible pass-through RJ45 crimper/stripper, used after a demonstration.
- One main + remote cable tester with a working battery.
- Teacher-prepared Cat5e practice cable, matching RJ45 plugs and boots.
- One known-good cable, this color card, pencils, a scrap cup and coded work tray.

Before students start

- Match plug rating to solid or stranded conductors, AWG, insulated-wire diameter and cable diameter.
- Match the tool to the pass-through plug; follow both manuals. Pilot a complete cable.
- Set stripping depth so no inner insulation is nicked. Pre-strip cable for students who need it.
- Demonstrate the pinch/blade zones, firm table support and flush tip inspection. Use eye protection for trimming.

Prepare enough for a real class

Start with **two short cables per table**, one per pair in a group of four. Each cable uses two new plugs and up to two boots. For seven tables, that is **14 cables, 28 plugs and 28 boots**, plus repair stock. Give each student an end-building turn where practical; record a turn still needed instead of rushing.

Example stock math, not a full-year promise: 350 plugs make at most 175 two-ended cables before mistakes. 200 ft of cable yields at most 100 two-foot blanks before trimming. Cable is the limit in that example. Crimpers, testers, boots and successful practice cables can be reused; used crimp plugs cannot. Scale cable and spare plugs to the actual number of classes and individual turns.

Two visits, with the same task at all seven tables

Minutes	Visit 1: understand + build	Visit 2: finish + troubleshoot
0-5	Sketch a local network. Trace a message. Name switch vs router.	Recall the route. Retrieve coded work and rotate jobs.
5-11	Demonstrate plug view, T568B, safe preparation and check-before-crimp.	Model two full tester runs. Compare a checked cable with a prepared fault.
11-26	Pairs build. Students sort/check; teacher circulates for approval. Record unfinished ends.	Finish, inspect and test. Log one observation, repair with a new plug if needed, and retest.
26-31	Explain one check. Label cable A/B and save actual progress.	Each student explains a network path and their own test observation.
31-35	Tools counted; wire scraps collected; work stored by class/table code.	Tester off; battery/tools checked; successful cables kept for future lessons.

What counts as success

Students explain a local connection, follow the correct pair order, and use evidence to keep or change their work. A wire-map pass alone is not cable certification. Do not connect student-made cables to school ports, powered network devices or PoE. The standalone tester supplies only its own test signal.

Test. Notice. Improve.

Class code: _____ Visit/date: _____ Record group codes only on shared sheets.

Check-in tracker · A and B are the two pair cables at each table

Table	Ends checked A/B	Wire map A/B	Change / next step	Turn still needed
1				
2				
3				
4				
5				
6				
7				

Troubleshooting at the table

We notice	Check next
No lights	Power, battery, correct ports and clicked-in plugs. Try the known-good cable before blaming the student cable.
A number is missing	Possible open connection: a wire may not reach the contact, or may be cut. Inspect both ends and replace the bad plug.
Numbers appear in the wrong order	Compare both ends against T568B with the plug held the same way. Replace the incorrectly ordered end.
Lights skip, combine or behave differently	Consult this tester's manual for its fault display. Inspect for touching tips, damaged insulation or poor contacts.
1-8 passes but something looks wrong	Inspect pair colors and twist, strain relief and flush tips. Cheap sequential testers can miss split pairs and cannot certify bandwidth.

Every student explains

1. Draw or say a path between two local devices.
2. What does a router do?
3. Which number or visual check gave you evidence?
4. What did you keep or change?

More support / more challenge

Support: enlarged numbered card, pre-stripped cable, partner grip, verbal or drawn record. Paper wire strips provide the same ordering task.

Stretch: diagnose a teacher-prepared misordered cable; explain why a 1-8 test does not measure speed.

Missing equipment? Draw devices, a switch and a router. Use eight numbered paper strips to model T568B at both ends. Follow each number across the cable and find a planted mistake. Students still explain a route, order and evidence. Record that no physical cable test occurred.

Teacher references: **Cisco: switch and router** · **TRENDnet: T568 wiring** · **trueCABLE: pair order and wire-map limits**. Follow the actual plug, crimper and tester manuals for operation.